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1. To show 256 colors, the number of bits required for each pixel is () ?

- A) 8 B) 16 C) 32 D) 64

2. If the boundary is specified in a single color, and if the algorithm proceeds pixel by pixel until the boundary color is encountered is called ().

- A) Scan-line fill algorithm
B) Boundary-fill algorithm
C) Flood-fill algorithm
D) Parallel curve algorithm

3. What are the RGB co-ordinates of a color at (0.25, 0.65, 0) in CMY space ? ()

- A) (0.25, 0.35, 0) B) (0.75, 0.65, 1) C) (0.75, 0.35, 0) D) (0.75, 0.35, 1)

4. Which code is not valid in Cohen-Sutherland line clipping algorithm? ()

- A) 1101 B) 1001 C) 0101 D) 0110

7. (多选) Which of the following is modern application for Computer Graphics? ()

- A) Computer Aided Geometric Design B) Video Games
C) Virtual Reality D) Scientific Visualization

8. (多选) Why do we need Mesh Simplification? ()

- A) Remove redundant geometry B) Reduce model size
C) Improve run-time performance D) Improve quality

9. (多选) Phong Lighting Model generally includes ().

- A) Ambient Light B) Diffuse Reflection
C) Specular Reflection D) Refraction

10. (多选) Projection can be categorized as ().

- A) Parallel Projection B) Oblique projection
C) Perspective Projection D) Orthographic projection

1. If the value of scaling factors S_x and S_y is greater than 1, then the size of objects will be enlarged. ()
2. The father of Computer Graphics is Ivan E. Sutherland. ()
3. 光线跟踪算法中包括了光反射和光透射光照效果。 ()
4. 走样指的是用离散量表示连续量引起的失真，是数字化的必然产物。 ()
5. In parallel projection, the distance between the projection center and the projection surface is finite. ()
6. Bezier 曲线两端点处切线方向必须与其特征多边形的相应两端线段走向一致。 ()

7. In Loop Subdivision, after n subdivision steps, each triangle will be subdivided into $3n$ triangles. ()

8. 在消隐算法中画家算法可以做到对所有场景进行隐藏面消除。 ()

9. 在镜面反射光中，镜面反射光的会聚指数（“高光”指数） n 越大，表面越粗糙。 ()

10. 在 X 扫描线填充算法中，为了避免填充扩大化，落在左、右边边界上的像素都不予填充。 ()

Short Answer Questions (4 items, 5 points each)

Score	
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1. What is the difference between graphics and image?
2. Describe the difference between window port (窗口) and view port (视口)?
3. What is mesh simplification and mesh subdivision?
4. What is the Gouraud Shading and Phong Shading. Write down the difference between Gouraud Shading and Phong Shading.

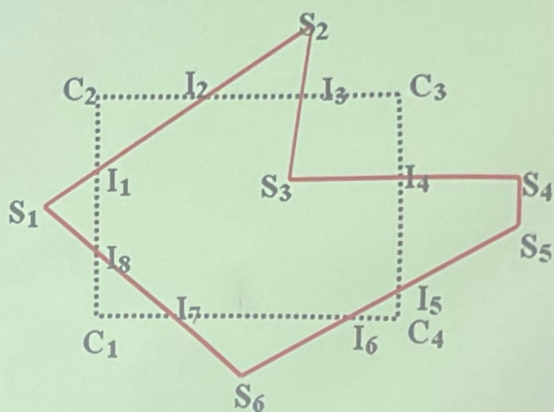
IV. Algorithms and Calculations (5 items, 10 points each)

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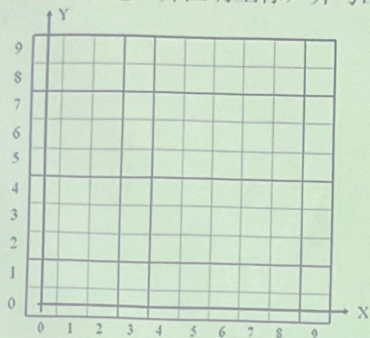
1. 证明关于 $y = -x$ 轴的对称变换等价于一个先对 y 轴做对称变换, 然后逆时针旋转 90 度的复合变换。



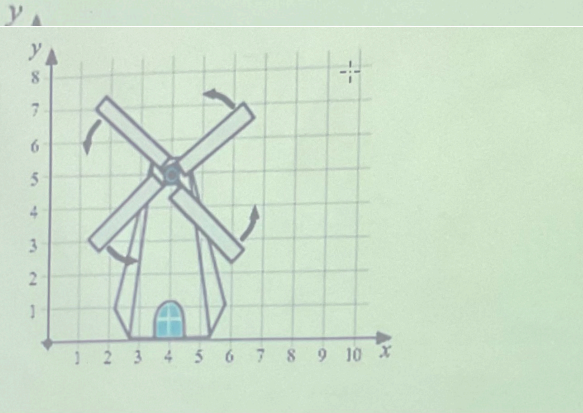
2. Clip the polygon using the Sutherland-Hodgeman algorithm. The input is $S_1 S_2 S_3 S_4 S_5 S_6$ and the order is Left, Top, Right, Bottom.



3. 用中点画线法 (中点 Bresenham 算法) 生成 $A(1, 1)$, $B(9, 5)$ 线段, 请画出其轨迹, 并注明坐标, 并写出计算过程。



4. A windmill is modeled in its final position in world coordinates as shown. Write a transformation that could be applied to just the windmill vanes (风车叶片), that will rotate them anticlockwise (逆时针) 45° so that they pivot about their axle as shown.
(请写出齐次坐标变换矩阵)



5. Construct Bezier curve of order 3, with 4 polygon vertices A(1, 1), B(2, 3), C(4, 3) D(6, 4) for values of u ($0 \leq u \leq 1$), where $p(u)$ is a point on curve with values for u , please calculate the value of $p(1/2)$.

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Score

I. Multiple Choices (5 items, 2 points each)

1. When a point $P(x, y)$ is rotated by θ the coordinates of transformed point P' are given as: ().

- A) $x' = x \cos(\theta) - y \sin(\theta), y' = x \sin(\theta) + y \cos(\theta)$
- B) $x' = y \cos(\theta) - x \sin(\theta), y' = y \sin(\theta) + x \cos(\theta)$
- C) $x' = x \cos(\theta) + y \sin(\theta), y' = x \sin(\theta) - y \cos(\theta)$
- D) $x' = y \cos(\theta) + x \sin(\theta), y' = y \sin(\theta) - x \cos(\theta)$

2. The highest award in computer graphics is named after ().

- A) Ivan E. Sutherland
- B) Pierre Bézier
- C) Steven A. Coons
- D) Bui-Tuong Phong

3. The Boundary Fill and Flood Fill algorithms: ().

- A) Must use 4-connected approach
- B) Must use 8-connected approach
- C) May use 4-connected or 8-connected approach
- D) Must not use recursive approach

4. The modern PC graphics card consists of three main components: ().

- A) GPU
- B) video memory
- C) RAMDAC
- D) ROM BIOS

5. Properties of good line drawing algorithm ().

- A) Line should appear straight
- B) Line should be drawn rapidly
- C) Lines should terminate accurately
- D) Lines should have constant density

II. Blank Fillings (5 items, 2 points each)

Score

1. In perspective projection, the maximum number of major vanishing points is _____.

2. X 扫描线算法中, 每次用一条扫描线进行填充, 对一条扫描线填充的过程可分为 4 个步骤: _____、_____、_____、_____。

3. 对于基本几何变换, 一般有平移、旋转、对称和错切等, 这些基本几何变换都是相对于_____和_____进行的几何变换。

4. 具有 256 级灰度、分辨率为 1024×1024 个像素阵列的光栅扫描式显示器需要_____KB 的缓冲器。

5. 利用中点画线法扫描转换直线 $P(0,0)$ 和 $P(5,2)$ 时, 最大的步长方向为: _____。

1. Both Boundary Filling and Flood filling algorithms are non-recursive techniques.

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2. To modify the object shape, shearing transformation cannot be used. ()

3. Using Cohen-Sutherland line clipping, it is impossible for a vertex to be labeled

1111.

()

4. Translation moves objects without deformation.

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5. Sutherland-Hodgeman clipping algorithm clips any polygon against a concave clip polygon.

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6. 字符的图形表示分为点阵和矢量两种形式。

()

7. 视区大小不变，当窗口变大时，映射到视区的物体也会相应变大。 ()

8. 使用齐次坐标可以将 n 维空间的一个点向量唯一映射到 $n+1$ 维空间中。()

9. 反走样技术可以完全消除走样现象。

()

10. LCD表示发光二极管显示器

()

IV. Short Answer Questions (5 items, 4 points each)

Score

1. What is the difference between computer graphics and image processing?

2. What do you mean by projection? Describe the difference between parallel projection and perspective projection.

3. Explain what is aliasing and what is antialiasing? Please list three antialiasing techniques.

功能: 不可用

4. Describe the 2D rotation transformation process with respect to any reference point.

5. 描述改进的一个深度缓存变量的 Z-buffer 算法的流程。

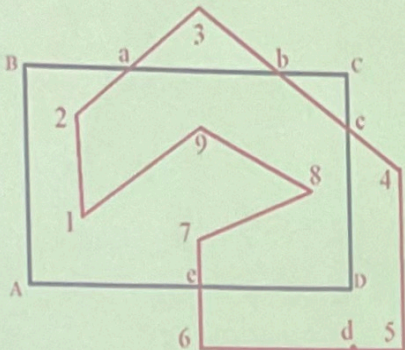
V. Algorithms and Calculations (5 items, 10 points each)

Score

1. Find the transformation matrix that reflects a polygon whose vertices are $A(-1, 0)$, $B(0, -2)$, $C(1, 0)$, and $D(0, 2)$ about the Y-axis, and write the normalized homogeneous coordinates of each vertex of the polygon after transformation.

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2. Clip the polygon using the Sutherland-Hodgeman algorithm. The Input is 123456789.



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